

1. In dental composite, the role of fillers is all the following except:
 - a) Reduce shrinkage upon curing.
 - b) Increase mechanical properties.
 - c) Decrease coefficient of thermal expansion.
 - d) Increase water sorption**
2. What is the primary disadvantage of using high molecular weight monomers (oligomers) like Bis-GMA in the organic matrix of a resin composite?
 - a) They cause high polymerization shrinkage.
 - b) Highly volatile during clinical use.
 - c) They possess low mechanical properties.
 - d) They have high viscosity limiting filler addition.**
3. When we use ceramic laminate veneer, the following cement can be used:
 - a) varnish
 - b) Calcium hydroxide
 - c) Zinc oxide eugenol cement
 - d) Resin cements**
4. Why does calcium hydroxide can be used as a pulp capping material?, due to:
 - a) Bonding chemically to tooth structure
 - b) Fluoride release
 - c) Its high strength
 - d) Stimulating reparative dentin**
- 5- We should manipulate wax (carving , shaping etc....) below the solid-solid transition temperature:
 - a) True
 - b) False**
- 6- ----- is used for etching ceramic laminate veneer:
 - a) Phosphoric acid 37%
 - b) Hydrofluoric acid 5-9%**
 - c) Polyacrylic acid 5%

d) HCl 20%

7- During fabrication of ceramo-metallic crown, porcelain slurry must wet metal core by contact angle:

- a) > 60 degrees
- b) 90 degrees
- c) < 60 degrees
- d) 75 degrees

8- A: Short term soft liners must be replaced at least every week.

B: They gain rubbery consistency by polymerization.

- a. A is only right
- b. B is only right
- c. Both A and B are right
- d. Both A and B are wrong

9- In dental composite, the role of fillers is all the following except:

- a) Reduce shrinkage upon curing.
- b) Increase mechanical properties.
- c) Increase coefficient of thermal expansion.
- d) Decrease water sorption

10-In composite filling,is used to bond filler with resin matrix

- a) Primer.
- b) Coupling agent.
- c) Adhesive.
- d) Acid etch

11- When we use all ceramic restoration, the following cement can be used:

- a) varnish
- b) Calcium hydroxide
- c) Zinc oxide eugenol cement
- d) Resin cements

12- Why does calcium hydroxide can be used as a pulp capping material?, due to:

- a) Bonding chemically to tooth structure
- b) Fluoride release
- c) Its high strength
- d) Stimulating reparative dentin

13- We should manipulate wax (carving , shaping etc....) in the solid-solid transition temperature:

- a. True
- b. False

14- ----- is used for etching ceramic laminate veneer:

- a) Phosphoric acid 37%
- b) Hydrofluoric acid 5-9%
- c) Polyacrylic acid 5%
- d) HCl 20%

15- During fabrication of ceramo-metallic crown, porcelain slurry must wet metal core by contact angle:

- a) > 60 degrees
- b) 90 degrees
- c) < 60 degrees
- d) 75 degrees

16- A: Short term soft liners must be replaced at least every 3 days.

B: They gain rubbery consistency by plasticizers.

- a. A is only right
- b. B is only right
- c. Both A and B are right
- d. Both A and B are wrong

17-Activating compound for heat cured PMMA is:

- a) Hydroquinone.
- b) Benzoyl peroxide
- c) Camphorquinone.

d) Potassium oxide.

18-TEGDMA monomers are added to composite resin to:

a) Decrease viscosity.

b) Increase viscosity.

c) Decrease shrinkage.

d) B&C.

19-Which of the following cements bonds chemically to tooth structure:

a) Zinc Polycarboxylate.

b) Resin cements.

c) Zinc oxide eugenol cement.

d) varnish

20-Regarding zinc phosphate cement:

a) Powder and liquid are mixed on cold glass slab.

b) Powder and liquid are mixed on hot glass slab.

c) Zinc phosphate cement has exothermic reaction.

d) a and c

21- Processing wax should have flow temperature:

a) At room temperature

b) Higher than room temperature

c) Lower than room temperature

d) At 200 ° C

22- High-fusing ceramic types are used for veneering ceramic-alloy and all-ceramic restorations:

a- True

b- False

23- Hard machining of all ceramic restorations needs:

- a) Fully sintered ceramic
- b) Partially sintered ceramic
- c) Milling of enlarged restorations
- d) b and c

24- A: Transformation toughening of 3Y-TZP is an extrinsic toughening mechanism.

B: It depends on transformation of monoclinic crystals to tetragonal upon crack propagation.

- a- A is only right
- b- B is only right
- c- Both A and B are right
- d- Both A and B are wrong

25-Activating compound for light cured PMMA is:

- e) Hydroquinone.
- f) Benzoyl peroxide
- g) Camphorquinone.
- h) Potassium oxide.

26-TEGDMA monomers are added to composite resin to:

- e) Decrease viscosity.
- f) Increase viscosity.
- g) Decrease shrinkage.
- h) B&C.

27-Which of the following cements bonds chemically to tooth structure, and has anti-cariogenic effect:

- e) Zinc Polycarboxylate.
- f) Resin cements.
- g) Zinc oxide eugenol cement.
- h) Glass ionomer cements

28-Regarding zinc phosphate cement:

- e) Powder and liquid are mixed on cold glass slab.
- f) Powder and liquid are mixed on hot glass slab.
- g) Zinc phosphate cement has endothermic reaction.
- h) a and c

29- Pattern wax should have flow temperature:

- a) At room temperature
- b) Higher than room temperature
- c) Lower than room temperature
- d) At 200 ° C

30- Low-fusing ceramic types are used for veneering ceramic–alloy and all-ceramic restorations:

- a- True
- b- False

31- Soft machining of all ceramic restorations needs:

- a) Fully sintered ceramic
- b) Partially sintered ceramic
- c) Milling of enlarged restorations
- d) b and c

32- A: Transformation toughening of 3Y-TZP is an extrinsic toughening mechanism.

B: It depends on transformation of monoclinic crystals to tetragonal upon crack propagation.

- a- A is only right
- b- B is only right
- c- Both A and B are right
- d- Both A and B are wrong

